Marcelo A. F. dos Santos

Ph.D. student at UC Davis mafsantos@ucdavis.edu

Research interests

High-energy physics, formal aspects of quantum field theory, string theory. I am interested in developing new technologies to help understand and compute scattering amplitudes. I have worked on geometrization of scattering amplitudes, pure spinor quantization, the worldline formalism, maximally supersymmetric theories, and analytical aspects of the S-matrix. Recently, I have become interested in the intersection of amplitudes and cosmology.

Education

2024–Ongoing	**Ph.D.** Physics, UC Davis, USA,	advisor: **Prof. Jaroslav Trnka**
2021-2023	**M.Sc.** Physics, IFT/ICTP-SAIFR, Brazil,	advisor: **Prof. Nathan Berkovits**
2021-2022	**M.Sc.** Physics, Perimeter Institute, Canada,	advisor: **Prof. Achim Kempf**
2017-2020	**B.Sc.** Physics, U. of São Paulo, Brazil,	advisor: **Prof. Gabriel Landi**

Publications

- *de Sitter Gluon Correlators in Momentum Space*
  with M. Navabi, G. Pimentel, F. Rost, J. Trnka (arXiv:2610.28724).
- *The 11D pure spinor ghost number zero vertex operator*
  with M. Guillen, E. Vianna (arXiv 2508.19744).
- *Tree-level 11D supergravity amplitudes from the pure spinor worldline*
  with M. Guillen, E. Vianna (arXiv 2508.19748). JHEP 01 (2026), 074.
- *The pure spinor superparticle and 10D SYM amplitudes*
  with M. Guillen, E. Vianna (arXiv 2508.19601). JHEP 12 (2025), 044.
- *Collisional thermometry for Gaussian systems*
  with G. Alves, G. Landi (arXiv 2405.12030). Phys.Rev.A 110 (2024) 5, 052421.

Talks

- *Combinatorics of the scaffolded superstring*
  at California Amplitudes, Spring 2026 (UC Davis).
- *11D SUGRA from the pure spinor worldline*
  at QCD meets Gravity Conference, 2025 (ICTP-SAIFR).
- *Generalizations of kinematics for loop integrands*
  at California Amplitudes, Fall 2025 (UCLA).
- *Superstring amplitudes from surfaces*
  at California Amplitudes, Spring 2025 (UC Davis).
- *Field theory amplitudes with pure spinor worldline*
  at California Amplitudes Spring 2025 (UC Davis).
- *Superstring amplitudes from surfaces*
  at IAS Amplitudes Meeting, March 2025.

Teaching

- Teaching Assistant, UC Davis, 2024-Ongoing.
- Teaching Assistant in the PSI Start program, Perimeter Institute, 2022.
- Private tutoring, 2020-Ongoing.

Awards and Fellowships

- Rod Reid Endowed Fellowship Award, UC Davis, 2026.
- UC Davis Graduate Program Fellowship, 2025.
- ICTP-SAIFR Young Physicist Fellowship, 2020.
- ICTP-SAIFR Young Physicist Award, 2019.
- Brazilian Mathematics Olympiad, Bronze Medal, 2018.